

(A) Know the laws on p. 95, 96 and 97.

(B) Know Theorem 1, p. 100

Examples: Do these, then check your answer with the solution given at the end of this section.

It is very important to write correctly.

① $\lim_{x \rightarrow 2} 3x^2 + 4 \sin \pi x + 5$

② $\lim_{x \rightarrow 3} \frac{x^2 + \sin(x)}{x}$

③ $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

④ $\lim_{t \rightarrow 0} \sqrt{t+4} - 2$

⑤ $\lim_{x \rightarrow 0} \frac{\sqrt{3+x} - \sqrt{3}}{x}$

⑥ $\lim_{x \rightarrow 0} |x|$

⑦ $\lim_{x \rightarrow -2} \frac{2 - |x|}{2 + x}$

⑧ $\lim_{x \rightarrow 0} \frac{\frac{1}{3+x} - \frac{1}{3}}{x}$

⑨ $\lim_{x \rightarrow 4} \frac{2 - \sqrt{x}}{4x - x^2}$

⑩ $\lim_{x \rightarrow 0} \frac{|x|}{x}$

(C) Theorem 2 (p. 101)

(D) Theorem 3 (p. 101) The Squeeze Theorem.

Examples

① Example 11 (p. 102)

② $\lim_{x \rightarrow 0^+} \sqrt{x} \arctan(x)$

③ $\lim_{x \rightarrow 0} x^2 \cos \frac{1}{x}$

Look out for:

$$-1 \leq \sin(x) \leq 1$$

$$-1 \leq \cos(x) \leq 1$$

$$-\frac{\pi}{2} < \arctan(x) < \frac{\pi}{2}$$

② $\lim_{x \rightarrow 0^+} \sqrt{x} \arctan(x)$

We know: $-\frac{\pi}{2} < \arctan x < \frac{\pi}{2}$

$$\Rightarrow -\frac{\pi}{2} \sqrt{x} < \sqrt{x} \arctan x < \frac{\pi}{2} \sqrt{x} \quad \forall x \geq 0$$

Now: $\lim_{x \rightarrow 0^+} -\frac{\pi}{2} \sqrt{x} = 0$

and $\lim_{x \rightarrow 0^+} \frac{\pi}{2} \sqrt{x} = 0$

$$\Rightarrow \lim_{x \rightarrow 0^+} \sqrt{x} \arctan(x) = 0, \text{ by the Squeeze Theorem.}$$

③ $\lim_{x \rightarrow 0} x^2 \cos \frac{1}{x}$

We know $-1 \leq \cos x \leq 1 \quad \forall x \in \mathbb{R} \setminus \{0\}$
 $\Rightarrow (-1)x^2 \leq x^2 \cos(x) \leq x^2$, since $x^2 \geq 0$

Now $\lim_{x \rightarrow 0} (-1)x^2 = 0$ and $\lim_{x \rightarrow 0} x^2 = 0$, and

it follows from the Squeeze Theorem that

$$\lim_{x \rightarrow 0} x^2 \cos x = 0.$$

(E) Special limits: see study guide

Examples: (Do these on your own, then compare with written out answers)

① $\lim_{x \rightarrow 0} \frac{\sin(4x)}{x}$

② $\lim_{x \rightarrow 2} \frac{\tan(x^2 - 4)}{x - 2}$

③ $\lim_{x \rightarrow 0} \frac{x^2}{\sin(x)}$

④ $\lim_{x \rightarrow 4} \frac{\sin(x - 4)}{x^2 - 16}$

⑤ $\lim_{x \rightarrow 4} \frac{\sin(x)}{x}$

Also:

⑥ $\lim_{x \rightarrow 0^+} \sqrt{x} e^{\sin(\pi/x)}$

⑦ $\lim_{x \rightarrow 0^-} \left(\frac{1}{x} - \frac{1}{|x|} \right)$



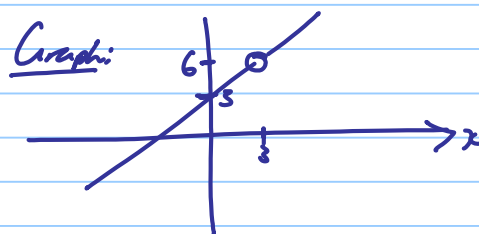
Solutions:

① $\lim_{x \rightarrow 2} 3x^2 + 4 \sin \pi x + 5 = 3(4) + 4 \sin(\pi \cdot 2) + 5$
 $= 12 + 4 + 5$
 $= 21.$

② $\lim_{x \rightarrow 3} \frac{x^2 + \sin(x)}{3} = \frac{9 + \sin(3)}{3}$

③ $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} \quad \left(\frac{0}{0} \right)$

$= \lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{x-3}$
 $= 6.$



④ $\lim_{t \rightarrow 0} \frac{t}{\sqrt{t+4} - 2} \quad \left(\frac{0}{0} \right)$
 $= \lim_{t \rightarrow 0} \frac{t(\sqrt{t+4} + 2)}{(\sqrt{t+4} - 2)(\sqrt{t+4} + 2)}$
 $= \lim_{t \rightarrow 0} \frac{t(\sqrt{t+4} + 2)}{t + 4 - 4}$

$$= \lim_{t \rightarrow 0} \frac{\cancel{t} (\sqrt{t+4} + 2)}{\cancel{t}}$$

$$= \sqrt{4} + 2 = \underline{4}$$

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

When x is close to -2 , $x < 0$
and $|x| = -x$.

$$(7) \lim_{x \rightarrow -2} \frac{2 - |x|}{2 + x}$$

$$= \lim_{x \rightarrow -2} \frac{2 - (-x)}{2 + x}$$

$$= \lim_{x \rightarrow -2} \frac{2 + x}{2 + x}$$

$$= \lim_{x \rightarrow -2} 1 = 1.$$

Further examples:

$$(1) \lim_{x \rightarrow 0} \frac{\sin(4x)}{x} = \lim_{x \rightarrow 0} \frac{4}{4} \frac{\sin(4x)}{x}$$

$$= 4 \lim_{x \rightarrow 0} \frac{\sin(4x)}{4x}$$

$$= 4 \times 1 = 4.$$

$$(2) \lim_{x \rightarrow 2} \frac{\tan(x^2 - 4)}{x - 2}$$

$$= \lim_{x \rightarrow 2} \frac{\tan(x-2)(x+2)}{x-2}$$

$$= \lim_{x \rightarrow 2} \frac{(x+2) \tan(x-2)(x+2)}{(x+2)(x-2)}$$

$$= \lim_{x \rightarrow 2} (x+2) \frac{\tan(x^2 - 4)}{x^2 - 4}$$

$$= \underbrace{\lim_{x \rightarrow 2} (x+2)}_4 \underbrace{\lim_{x \rightarrow 2} \frac{\tan(x^2 - 4)}{x^2 - 4}}_1$$

$$= \underline{4}$$

$$(6) \lim_{x \rightarrow 0^+} \sqrt{x} e^{\sin(\pi/x)}$$

$$-1 \leq \sin(\pi/x) \leq 1$$

$$\Rightarrow e^{-1} \leq e^{\sin(\pi/x)} \leq e^1$$

$$\Rightarrow \frac{\sqrt{x}}{e} \leq \sqrt{x} e^{\sin(\pi/x)} \leq \sqrt{x} e$$

$$\text{Now: } \lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{e} = 0 \quad \text{and} \quad \lim_{x \rightarrow 0^+} \sqrt{x} e = 0 \quad \Rightarrow \text{By the Squeeze Theorem, } \lim_{x \rightarrow 0^+} \sqrt{x} e^{\sin(\pi/x)} = 0$$

